

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network
Security **COURSE CODE:** CSA5116

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5

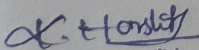
Total Marks:50

Annexure A

CONCEPT MAPPING

Code of Conduct:

I K. Harshitha (192324202) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.


Signature of the student:

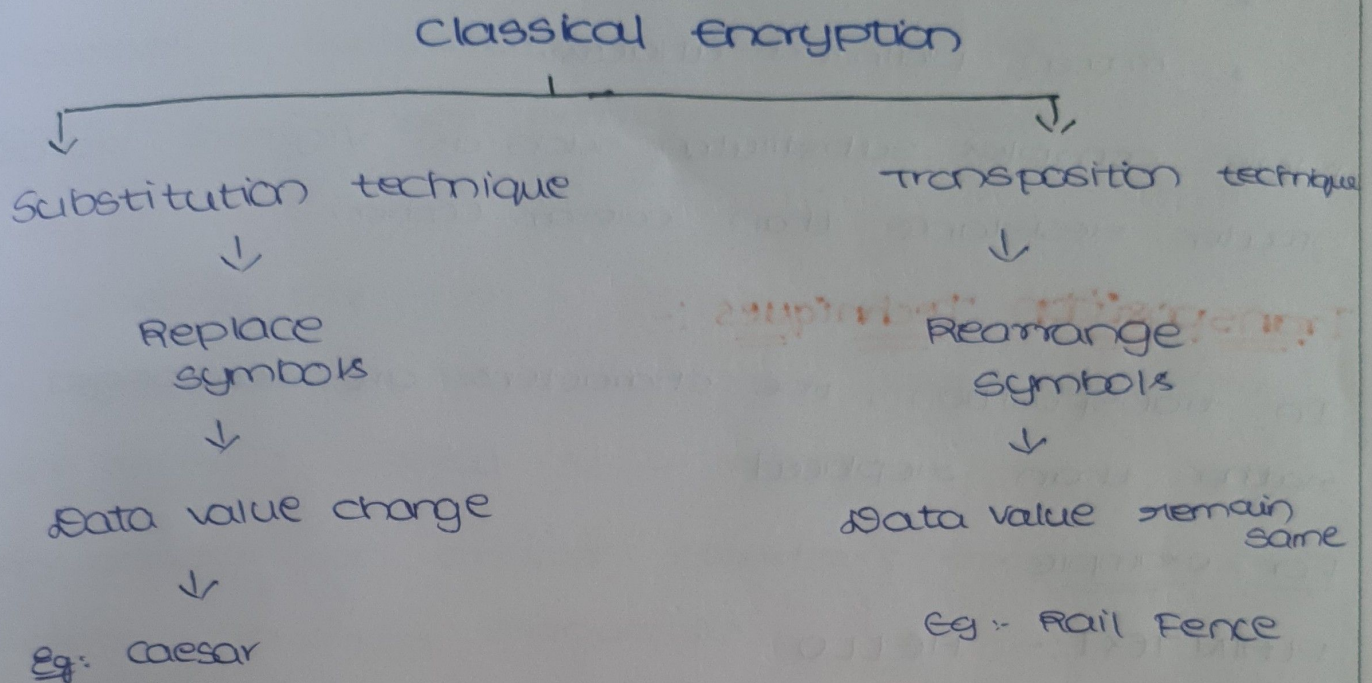
CONCEPT MAPPING

1. Create a concept map linking the following: Security Attacks, Threats, Risks and Security Goals (Confidentiality, Integrity, Availability). Perform the following tasks:
 - a. Show how each attack leads to specific threats and risks
 - b. Map how security goals help mitigate them Provide one real-world example for each link
2. Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks to connect based on: Key usage, Complexity and Security strength. Analyze which technique is most secure and why.
3. Construct a concept map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques. Perform the following tasks:
 - a. Show differences and similarities
 - i. Map how each affects the Data structure and Security level
 - ii. Analyze which is more vulnerable to attacks and why
4. Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:
 - a. Show relationships between these concepts
 - b. Analyze how each contributes to secure communication
 - c. Include one real-world application for each
5. Draw a concept map linking number theory concepts used in cryptography: Modular Inverse, Extended Euclid Algorithm, Fermat's Little Theorem, Euler's Phi Function and Euler's Theorem. Perform the following tasks:
 - a. Show how each concept is mathematically connected
 - b. Map their role in encryption/decryption
 - c. Analyze which concept is most critical for public key cryptography and justify

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding ; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

Construct a concept map for classical encryption schemes :- Substitution and Transposition Technique.
Classical encryption techniques are mainly divided into:



a) Differences and similarities

(i) How each affects data structure and security level

Substitution techniques :-

In substitution one character is replaced by another

Eg:- Plaintext :- HELLO

Ciphertext :- KHOOR

So,

- Character position remain the same.
- Number of characters remains the same.
- Character identities change.

Security :-

Simple substitution methods such as Caesar

Cipher are relatively weak because attackers can use:

- Frequency analysis
- Brute-force attacks
- Known plaintext attacks.

More complex substitution such as Vigenere provide greater resistance than caesar cipher.

Transposition Techniques :-

In transposition, the characters are rearranged rather than replaced.

For example :-

PLAIN TEXT :- HELLO

After transposition :- LHOEL

The exact result depends on transposition method & ^{key}.

(i) Map how each effects the data structure & security
Security:- Simple transposition techniques are also vulnerable because the attacker can analyse:

- character frequencies
- Repeated patterns
- possible arrangements
- Likely word structures.

However, a sufficiently complex transposition scheme can make reconstruction more difficult.

which is more vulnerable to attacks and why?

There is no universal answer that one entire category is always more vulnerable. It depends on the particular algorithm and key size.

However, a substitution techniques such as Caesar cipher are particularly vulnerable

why?

consider:- plaintext :- HELLO WORLD

In substitution, the frequency pattern of letters can remain useful to an attacker.

→ An attacker can compare ciphertext frequencies with expected language frequencies.

Transposition weakness :-

Transposition preserves the original character frequencies.

Overall Analysis :- Simple substitution :- vulnerable because repeated characters and language frequency can reveal pattern.

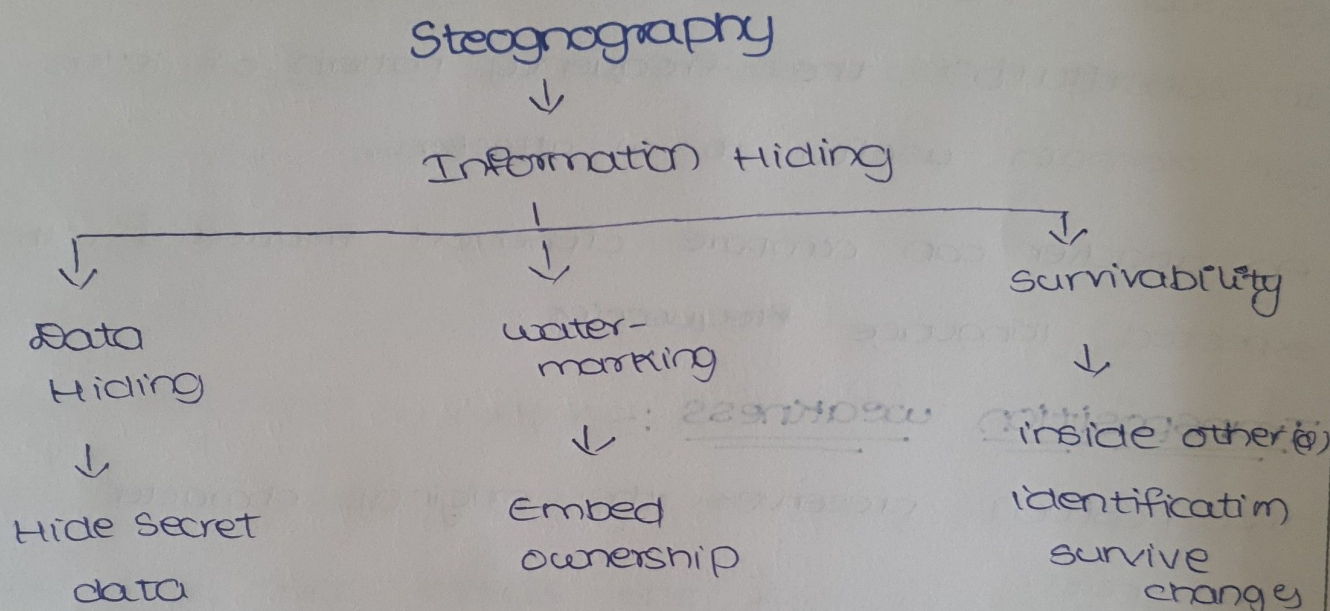
Transposition :- vulnerable because the exact character frequencies are preserved and possible arrangements can be tested.

4) Concept Map :- steganography

The main concepts are :-

- Data Hiding
- Watermarking
- Survivability

They are related to the broader concept of steganography and information hiding.



① Relationship between Data Hiding, watermarking and survivability.

1) Data Hiding :-

Data Hiding means embedding secret information inside another carrier.

The carrier could be:

- Image
- Video
- Audio
- Document

3.1- Secret message + Image



Stego Image

The image looks normal, but it contains additional hidden information.

Main goal :-

The main goal is to hide the existence of the secret information.

a) Watermarking :- watermarking is a specific application of information hiding where information is embedded.

b) How each contributes to secure communication.

Data Hiding → confidentiality and covert communication

Instead of sending → CONFIDENTIAL DATA

secret-data



Data Hiding



Normal looking image.

Its security contribution is mainly concealing the existence of communication.

Watermarking :- Authenticity and ownership

→ watermarking can help establish:

- Ownership
- Authenticity
- Integrity
- Copyright information.

Survivability :- Reliability and Persistence

Survivability makes the embedded information more reliable.

For example :-

watermark
↓

Image compression → Image resizing → mark remains.

c) One - real world application for each.

1) Data Hiding :-

Application :- convert communication in digital images

For eg :- confidential information

↓
data hiding → normal image → communication

Purpose :- convert communication

2) Watermarking :-

Application :- Digital photography and online media

For eg :- photograph + copyright information

↓
watermarked photograph

Purpose :- Copyright and ownership protection.

3) Survivability :-

Application :- Video streaming and digital media distribution.

Purpose :- Ensure that ownership (&) identification information.